

# Data Tracking Plan (Cocktail Recipes MVP) V1

## 1.General information

Product: Cocktail Recipes Website

Goal: Analysis of searches, filter usage and cocktail popularity

## 2.The system should record the following events (event tracking)

| No/n | Event              | The meaning                   |
|------|--------------------|-------------------------------|
| 1    | age_confirmation   | user confirmed / rejected 18+ |
| 2    | page_view          | open page                     |
| 3    | search_query       | text search                   |
| 4    | filter_applied     | apply filter                  |
| 5    | filters_reset      | reset filters                 |
| 6    | cocktail_card_view | view card                     |
| 7    | cocktail_page_open | open cocktail page            |
| 8    | servings_changed   | change number of servings     |

### 2.1 Age Confirmation

| Field       | Value                          |
|-------------|--------------------------------|
| Event name  | age_confirmation               |
| Description | user confirmed or rejected 18+ |
| Trigger     | clicking the button Yes / No   |

### Parameters

| Parameter     | Type     | Description        |
|---------------|----------|--------------------|
| age_confirmed | boolean  | true/false         |
| timestamp     | datetime | event time         |
| session_id    | string   | session identifier |

## 2.2 Page View

| Field       | Value            |
|-------------|------------------|
| Event name  | page_view        |
| Description | page view        |
| Trigger     | any page opening |

### Parameters

| Parameter  | Type   | Description                  |
|------------|--------|------------------------------|
| page_name  | string | home / search/ cocktail_page |
| page_url   | string | page URL                     |
| session_id | string | session identifier           |

## 2.3 Search Query

| Field       | Value                         |
|-------------|-------------------------------|
| Event name  | search_query                  |
| Description | text search for cocktails     |
| Trigger     | user clicks the search button |

### Parameters

| Parameter   | Type   | Description  |
|-------------|--------|--------------|
| search_text | string | request text |

|                        |         |                               |
|------------------------|---------|-------------------------------|
| <b>results_count</b>   | integer | result count                  |
| <b>filters_applied</b> | boolean | whether the filters were used |

## 2.4 Filter Applied

| Field              | Value                       |
|--------------------|-----------------------------|
| <b>Event name</b>  | filter_applied              |
| <b>Description</b> | the user applied the filter |

### Parameters

| Parameter            | Type    | Description                        |
|----------------------|---------|------------------------------------|
| <b>filter_type</b>   | string  | alcohol / sweetness / price / vibe |
| <b>filter_value</b>  | string  | filter value                       |
| <b>results_count</b> | integer | found cocktails count              |

## 2.5 Reset Filters

| Field              | Value                      |
|--------------------|----------------------------|
| <b>Event name</b>  | filters_reset              |
| <b>Description</b> | the user reset the filters |

### Parameters

| Parameter               | Type   | Description  |
|-------------------------|--------|--------------|
| <b>previous_filters</b> | string | filters list |

## 2.6 Cocktail Card View

| Field | Value |
|-------|-------|
|-------|-------|

|                    |                                  |
|--------------------|----------------------------------|
| <b>Event name</b>  | cocktail_card_view               |
| <b>Description</b> | cocktail shown in search results |

### Parameters

| Parameter   | Type    | Description          |
|-------------|---------|----------------------|
| cocktail_id | string  | cocktail ID          |
| position    | integer | position in the list |

## 2.7 Cocktail Page Open

| Field              | Value                 |
|--------------------|-----------------------|
| <b>Event name</b>  | cocktail_page_open    |
| <b>Description</b> | cocktail page opening |

### Parameters

| Parameter   | Type   | Description         |
|-------------|--------|---------------------|
| cocktail_id | string | ID                  |
| position    | int    | position in results |

## 2.8 Servings Calculator

| Field              | Value                     |
|--------------------|---------------------------|
| <b>Event name</b>  | servings_changed          |
| <b>Description</b> | change number of servings |

### Parameters

| Parameter   | Type   | Description |
|-------------|--------|-------------|
| cocktail_id | string | cocktail    |

|                        |         |                |
|------------------------|---------|----------------|
| <b>servings_number</b> | integer | portion number |
|------------------------|---------|----------------|

### 3. Filter reference values

#### **Alcohol Type**

vodka  
rum  
gin  
tequila  
whiskey  
liqueur  
non-alcoholic

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#### **Alcohol Level**

low  
medium  
strong

---

#### **Sweetness Level**

dry  
medium  
sweet

---

#### **Price Range**

Filter by price range: from 0 to max price

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#### **Vibe**

party  
romantic  
summer  
tropical

classic

---

## 4. Key product metrics

### Search Metrics

- total searches
  - searches with results
  - zero-result searches
  - top search queries
- 

### Filter Metrics

- most used filters
  - most used filter combinations
- 

### Cocktail Metrics

- most viewed cocktails
- cocktail CTR

$\text{cocktail\_page\_open} / \text{cocktail\_card\_view}$

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### Calculator Usage

$\text{servings\_changed} / \text{cocktail\_page\_open}$

### Conversion

$\text{search\_query} \rightarrow \text{cocktail\_page\_open}$

## 5. Analytics questions Examples

1. What cocktails are searched for most often?
2. What filters are used most often?
3. What search queries do not return results?
4. What cocktails are opened most often after a search?

5. Do users use a portion calculator?

## 6. Database Scheme

### 1. Cocktails Table

| Column          | Type     | Description           |
|-----------------|----------|-----------------------|
| cocktail_id     | int (PK) | unique cocktail ID    |
| name            | varchar  | cocktail name         |
| description     | text     | description           |
| alcohol_level   | varchar  | low / medium / strong |
| sweetness_level | varchar  | dry / medium / sweet  |
| price_usd       | decimal  | average price         |
| preparation     | text     | cooking method        |
| image_url       | varchar  | photo                 |

|                   |                  |                      |
|-------------------|------------------|----------------------|
| <b>created_at</b> | <b>timestamp</b> | <b>creation date</b> |
|-------------------|------------------|----------------------|

## 2. Ingredients Table

| <b>column</b>          | <b>type</b>                                | <b>description</b> |
|------------------------|--------------------------------------------|--------------------|
| <b>ingredient_id</b>   | <b>int (PK)</b>                            |                    |
| <b>ingredient_name</b> | <b>varchar</b>                             |                    |
| <b>category</b>        | <b>varchar (alcohol / mixer / garnish)</b> |                    |
| <b>unit</b>            | <b>varchar</b>                             |                    |

This is the ingredient master data.

It contains a unique list of all ingredients that exist in the system.

### Example

| <b>ingredient_id</b> | <b>ingredient_name</b> | <b>category</b> |
|----------------------|------------------------|-----------------|
| <b>1</b>             | <b>Vodka</b>           | <b>alcohol</b>  |
| <b>2</b>             | <b>Lime juice</b>      | <b>mixer</b>    |



|          |                    |                |
|----------|--------------------|----------------|
| <b>3</b> | <b>Mint</b>        | <b>garnish</b> |
| <b>4</b> | <b>Sugar syrup</b> | <b>mixer</b>   |

One record = one ingredient.

### 3. Cocktail\_ingredients Table

Since one cocktail has many ingredients, a many-to-many table is needed.

| <b>Column</b>        | <b>Type</b>     |
|----------------------|-----------------|
| <b>id</b>            | <b>int (PK)</b> |
| <b>cocktail_id</b>   | <b>int (FK)</b> |
| <b>ingredient_id</b> | <b>int (FK)</b> |
| <b>amount</b>        | <b>decimal</b>  |

This is a correlation table + recipe.

It shows:

- which ingredient
- goes into which cocktail
- in what quantity

**Example:**

| <b>id</b> | <b>cocktail_id</b> | <b>ingredient_id</b> | <b>amount</b> | <b>unit</b>   |
|-----------|--------------------|----------------------|---------------|---------------|
| <b>1</b>  | <b>10</b>          | <b>1</b>             | <b>50</b>     | <b>ml</b>     |
| <b>2</b>  | <b>10</b>          | <b>2</b>             | <b>25</b>     | <b>ml</b>     |
| <b>3</b>  | <b>10</b>          | <b>3</b>             | <b>5</b>      | <b>leaves</b> |

**This means:**

**Cocktail 10 (e.g. Mojito) includes:**

- **Vodka**
- **Lime juice**
- **Mint**

## **4. Tags Table**

| <b>Column</b>   | <b>Type</b>     |
|-----------------|-----------------|
| <b>tag_id</b>   | <b>int (PK)</b> |
| <b>tag_name</b> | <b>varchar</b>  |

**Example:**

- **summer**

- `tropical`
- `party`
- `classic`

## 5. Cocktail\_tags Table

Many-to-many relationship.

| column                   | type                  |
|--------------------------|-----------------------|
| <code>id</code>          | <code>int (PK)</code> |
| <code>cocktail_id</code> | <code>int (FK)</code> |
| <code>tag_id</code>      | <code>int (FK)</code> |

## 6. Cocktail\_similar Table

For block "similar cocktails".

| column                   | type             |
|--------------------------|------------------|
| <code>id</code>          | <code>int</code> |
| <code>cocktail_id</code> | <code>int</code> |

|                            |            |
|----------------------------|------------|
| <b>similar_cocktail_id</b> | <b>int</b> |
|----------------------------|------------|

## 7. Events Table (for Data Analysis)

| <b>column</b>            | <b>type</b>      | <b>description</b>               |
|--------------------------|------------------|----------------------------------|
| <b>event_id</b>          | <b>int (PK)</b>  | <b>event id</b>                  |
| <b>event_name</b>        | <b>varchar</b>   | <b>event type</b>                |
| <b>timestamp</b>         | <b>timestamp</b> | <b>event time</b>                |
| <b>anonymous_user_id</b> | <b>varchar</b>   | <b>anonymous user id</b>         |
| <b>session_id</b>        | <b>varchar</b>   | <b>session id</b>                |
| <b>page_name</b>         | <b>varchar</b>   | <b>home/search/cocktail page</b> |
| <b>cocktail_id</b>       | <b>int</b>       | <b>cocktail id</b>               |
| <b>search_text</b>       | <b>varchar</b>   | <b>search query</b>              |

|                        |                |                            |
|------------------------|----------------|----------------------------|
| <b>filter_type</b>     | <b>varchar</b> | <b>filter type</b>         |
| <b>filter_value</b>    | <b>varchar</b> | <b>filter value</b>        |
| <b>results_count</b>   | <b>int</b>     | <b>number of results</b>   |
| <b>servings_number</b> | <b>int</b>     | <b>number of servings</b>  |
| <b>age_confirmed</b>   | <b>boolean</b> | <b>age confirmation</b>    |
| <b>position</b>        | <b>int</b>     | <b>position in results</b> |

## 8. Sessions Table

| <b>column</b>            | <b>type</b>         |
|--------------------------|---------------------|
| <b>session_id</b>        | <b>varchar (PK)</b> |
| <b>anonymous_user_id</b> | <b>varchar</b>      |
| <b>session_start</b>     | <b>timestamp</b>    |
| <b>session_end</b>       | <b>timestamp</b>    |

|             |         |
|-------------|---------|
| device_type | varchar |
| browser     | varchar |

## 9. Page\_views Table

| column      | type      |
|-------------|-----------|
| id          | bigint    |
| session_id  | varchar   |
| page_name   | varchar   |
| cocktail_id | int       |
| timestamp   | timestamp |

## 7. Intire list of the relations

### ERD

### Cardinality

| Від → До | Type | Optional / Mandatory | Explanation |
|----------|------|----------------------|-------------|
|----------|------|----------------------|-------------|

|                                    |          |                |                                                                                                      |
|------------------------------------|----------|----------------|------------------------------------------------------------------------------------------------------|
| Cocktails → Cocktail_ingredients   | 1 → many | mandatory 1..* | A cocktail always has ≥1 ingredients                                                                 |
| Cocktail_ingredients → Cocktails   | many → 1 | mandatory 1..1 | 1 and only 1                                                                                         |
| Ingredients → Cocktail_ingredients | 1 → many | optional 0..*  | A single ingredient may not be used in any cocktail                                                  |
| Cocktail_ingredients → Ingredients | many → 1 | mandatory 1..1 | Each recipe line must have exactly one ingredient                                                    |
| Cocktails → Cocktail_tags          | 1 → many | optional 0..*  | A cocktail can have from 0 to many tags, 0..many, because a cocktail can be untagged<br>1 and only 1 |
| Cocktail_tags → Cocktails          | many → 1 | mandatory 1..1 | 1 and only 1                                                                                         |
| Tags → Cocktail_tags               | 1 → many | optional 0..*  | 0 to many, i.e. a tag may not belong to any cocktail_tags                                            |
| Cocktail_tags → Tags               | many → 1 | mandatory 1..1 | 1 and only 1                                                                                         |
| Cocktails → Cocktail_similar       | 1 → many | optional 0..*  | 0 to many, i.e. a cocktail can have 0 or many similar ones                                           |
| Cocktail_similar → Cocktails       | many → 1 | mandatory 1..1 | 1 and only 1 – each connection line always refers to a specific cocktail                             |
| Sessions → Events                  | 1 → many | optional 0..*  | a session can have 0 to many events                                                                  |
| Events → Sessions                  | many → 1 | mandatory 1..1 | 1 and only 1                                                                                         |
| Sessions → Page_views              | 1 → many | optional 0..*  | a session can have 0 or many page views                                                              |
| Page_views → Sessions              | many → 1 | mandatory 1..1 | 1 and only 1                                                                                         |
| Cocktails → Events                 | 1 → many | optional 0..*  | a cocktail can have 0 or many events                                                                 |

|                        |          |               |                                                                                                          |
|------------------------|----------|---------------|----------------------------------------------------------------------------------------------------------|
| Events → Cocktails     | many → 1 | optional 0..1 | 1 and only 1 – each event is related to a specific cocktail or null if the event is not about a cocktail |
| Cocktails → Page_views | 1 → many | optional 0..* | a cocktail can have 0 or many page views                                                                 |
| Page_views → Cocktails | many → 1 | optional 0..1 | 1 and only 1                                                                                             |